

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Define C₄ Compounds. Describe the manufacturing process of phenol from cumene with a clean flow sheet and all the necessary reactions. (CO4)

Q.24 Write a short note on any two of the following-

- a) Non-Regenerative Fixed Bed Process (CO3)
- b) Atmospheric Distillation of crude oil (CO2)
- c) Composition and classification of petroleum (CO1)

Q.25 Explain the manufacturing of formaldehyde with a neat flow sheet and state its uses and properties. (CO4)

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Roll No.

5th Sem/ Chemical

Subject : Petroleum Refining & Petrochemicals

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which method is used for removal of salt from crude oil (CO1)

- a) Dehydration of crude oil
- b) Stabilization of crude oil
- c) Electrical desalting of crude oil
- d) None of these

Q.2 Tetraethyl Lead is added to gasoline to (CO2)

- a) Reduce gum formation
- b) Increase smoke point
- c) Reduce the pour point
- d) Increase Octane Number

Q.3 The main aim of Reforming is to produce (CO3)

- a) Gasoline
- b) Lube Oil
- c) Coke
- d) None of these

- Q.4 How can we minimize the formation of Poly alkyl benzene in the manufacturing of cumene? (CO4)
- Using High Benzene and Propylene feed
 - Using high-end catalyst
 - Using low end catalyst
 - Using low Benzene and Propylene feed
- Q.5 What is the molecular weight of phenol? (CO4)
- 44
 - 94
 - 84
 - 48
- Q.6 Cyclohexane (C_6H_{12}) is related to (CO1)
- Paraffins
 - Aromatic
 - Naphthenes
 - Olefins

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Write the name of refinery in India having largest Capacity. (CO1)
- Q.8 Cetane number is the property of ____ (Diesel / Petrol) (CO2)
- Q.9 Name any two products obtained from fractional distillation of crude oil. (CO2)
- Q.10 Name any one type of reforming process. (CO3)

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- Q.11 List one important petrochemical derived from C_1 compound. (CO4)
- Q.12 Write the chemical formula of styrene. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss the origin of petroleum in brief. (CO1)
- Q.14 Enlist any four petroleum products with their major end uses. (CO2)
- Q.15 Explain the Concept of Octane and Cetane number. (CO2)
- Q.16 Write a short note on transportation of crude oil. (CO1)
- Q.17 State four uses of ethanol. (CO4)
- Q.18 Explain cracking and its types. (CO3)
- Q.19 With the help of flow sheet briefly describe the manufacturing of butadiene. (CO4)
- Q.20 Describe electrical desalting of crude oil with neat & clean diagram. (CO1)
- Q.21 Discuss the chemical reactions involved in catalytic Reforming. (CO3)
- Q.22 Differentiate between cloud point and pour point. (CO2)

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